

MARKET INTELLIGENCE PROJECT

Data Methodology Report

Cloud Kitchen Intelligence Study

College Road, Nashik · Swiggy / Zomato Data

Project Title	Cloud Kitchen Market Intelligence – College Road, Nashik
Data Sources	Swiggy / Zomato (manual extraction via browser inspection)
Scope	35 restaurants · 5 deep-dive menus · 1 locality
Report Type	End-to-End Data Methodology & Business Insight Report
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1. Introduction

1.1 Project Objective

This report documents the complete data workflow undertaken for the Cloud Kitchen Market Intelligence project, covering College Road, Nashik. The primary objective was to collect, clean, and analyse restaurant data from major food delivery platforms — Swiggy and Zomato — to identify market gaps, pricing trends, and viable opportunities for launching a cloud kitchen in the selected locality.

1.2 Relevance of Cloud Kitchen Intelligence

The Indian food delivery market has undergone significant disruption over the past decade. Cloud kitchens — delivery-only restaurant operations without a dine-in setup — represent a growing segment that benefits from lower operating costs and higher margin potential. However, success in this model depends critically on data: understanding which cuisines are already saturated, which price points attract repeat orders, and what operational patterns distinguish high-performing outlets from average ones.

Market intelligence derived from existing platform data provides a cost-effective, evidence-based foundation for such decisions before a single rupee of capital is deployed.

1.3 Scope

The study is geographically scoped to College Road, Nashik — a high-density corridor with a significant student and working-professional population. The dataset covers:

- 35 restaurants listed on Swiggy/Zomato as of the data collection date
- 5 selected restaurants subjected to detailed menu-level analysis
- 1 relational database schema designed for ongoing intelligence queries
- 4 core business intelligence dimensions: cuisine saturation, pricing, operational efficiency, and market gap identification

2. Data Collection Process

2.1 Source Platforms

Data was sourced from two primary food delivery platforms:

- **Swiggy:** Used as the primary source for restaurant listing data including ratings, review counts, delivery estimates, and cuisine tags.
- **Zomato:** Used as a secondary validation source for menu pricing, bestseller identification, and restaurant type classification.

2.2 Collection Methodology

Data was collected through a structured manual extraction process using browser developer tools (Network tab inspection). The process involved:

- Navigating to the College Road, Nashik locality feed on both platforms
- Inspecting XHR/Fetch network calls to identify the underlying API response payloads (JSON format)
- Extracting relevant fields from the JSON structure for each restaurant listing
- Recording data into a working spreadsheet in real time for 30–40 restaurant records

For the 5 deep-dive restaurants (KFC, McDonald's, Pizza Hut, Oven Story Pizza, Theobroma), menu data was extracted separately by:

- Opening each restaurant's menu page on the delivery platform
- Manually recording category names, item names, and listed prices

- Noting platform-flagged "Popular" or "Recommended" indicators as proxies for bestseller status

2.3 Fields Collected

Restaurant Name	Primary identifier — exact name as listed on the platform
Rating	Decimal rating (0.0–5.0 scale)
Number of Reviews	Integer count of customer reviews
Cost for Two	Approximate cost in INR for two people (platform-provided estimate)
Cuisines	Comma-separated tags (e.g., "North Indian, Chinese, Biryani")
Locality	Listed locality — all records confirmed as College Road
Restaurant Type	Classified as Cloud Kitchen, Dine-In, or Unknown
Delivery Time	String range as listed on platform (e.g., "30–35 min")

3. Tools and Platforms Used

Browser DevTools	Chrome DevTools (Network tab) used for API payload inspection and raw JSON extraction from Swiggy/Zomato feeds
Microsoft Excel	Primary workspace for raw data entry, initial structuring, and quick visual inspection of collected records
Python (pandas)	Used for scripted data cleaning, cuisine string parsing, type normalization, and NULL handling logic
MySQL 8.0	Used to design the normalised relational schema and write analytical SQL queries (Q1–Q6)
openpyxl / xlswriter	Used to generate multi-sheet Excel output files with cleaning report summaries embedded
CSV (plain text)	Intermediate format for portability between tools and for final cleaned dataset delivery

4. Data Cleaning and Transformation

4.1 Missing Value Handling

Missing values were identified across four fields. The handling strategy for each was chosen to preserve analytical integrity rather than artificially inflate data quality:

- **Rating (6 records missing):** Retained as NULL. Imputing an average rating would misrepresent outlet quality and distort ranking queries.
- **Number of Reviews (6 records missing):** Retained as NULL. Filling with 0 would imply zero customer activity, which is factually unverified.
- **Cost for Two (6 records missing):** Retained as NULL. No reliable proxy existed for estimation without introducing bias.

- **Delivery Time (21 records missing):** Retained as NULL. Platform may not have displayed delivery estimates for all outlets; no substitution was made.

4.2 Duplicate Detection

Duplicate records were identified using a composite key of Restaurant Name + Locality. No duplicate records were found across the 35-record dataset. All restaurant names were verified to be distinct within the College Road locality scope.

4.3 Cuisine Normalization

Cuisine tags represented the most inconsistent field in the raw data, with multiple variants referring to the same category. The following normalization rules were applied:

- "north-indian", "north indian food", "North Indian cuisine" → North Indian
- "fast-food", "fastfood", "Fast food" → Fast Food
- "middle-eastern", "Middle Eastern food" → Middle Eastern
- "south-indian", "South Indian" → South Indian

Normalization was applied using regex-based pattern matching. Unmatched cuisine strings were title-cased as-is to preserve information without distortion.

4.4 Restaurant Type Standardization

The raw data contained inconsistent type labels. All records were normalized to one of three permitted values:

- Cloud Kitchen
- Dine-In
- Unknown (for records where type could not be reliably determined)

One record (Pizza Hut) was listed as "Cloud + Dine-in" — a dual classification not supported by the schema. This record was reclassified as Cloud Kitchen, as Pizza Hut College Road primarily serves delivery orders with no dedicated dine-in floor in this outlet type.

4.5 Numeric Field Formatting

- **Cost for Two:** The ₹ symbol and commas were stripped. Values converted to integer type.
- **Rating:** Stored as DECIMAL(2,1) — e.g., 4 converted to 4.0 for consistent formatting.
- **Number of Reviews:** Commas stripped (e.g., "1,200" → 1200). No K-notation values were found.
- **Delivery Time:** Split into two integer columns (delivery_time_min, delivery_time_max) in the database schema. Raw string format retained in the flat CSV.

5. Assumptions Made

- Missing values (rating, reviews, cost, delivery time) were assumed to reflect absent platform data — not zero values — and were treated as NULL throughout.

- Multi-cuisine tags were treated as equal-weight descriptors. A restaurant listed as "Pizza, Italian" was counted once each for Pizza and Italian in saturation analysis.
- Delivery time strings extracted from the platform were accepted as stated and not independently verified. Values reflect platform estimates, not actual delivery performance.
- Bestseller identification for menu items was based on three signals: platform-flagged indicators ("Popular", "Recommended"), menu positioning (top-listed items), and known brand signature products. These are approximations, not sales data.
- All 35 records were assumed to represent active restaurants operational at the time of data collection. Temporarily closed or recently delisted outlets may not have been captured.
- Menu prices recorded represent listed delivery prices and may differ from dine-in pricing or discounted offer prices active during data collection.

6. Challenges Faced

6.1 Dynamic Platform Rendering

Both Swiggy and Zomato use JavaScript-heavy, dynamically loaded interfaces. Restaurant listings are not embedded in static HTML — they are fetched via XHR/fetch calls after initial page load. This required using browser DevTools to intercept API payloads rather than relying on standard HTML parsing methods.

6.2 Inconsistent Cuisine Formatting

Cuisine tags varied significantly across listings — including hyphenated forms, lowercase variants, redundant suffixes ("food", "cuisine"), and non-standard names. Constructing a regex normalization map required manual inspection of all unique cuisine values before automation could be applied.

6.3 Incomplete Review and Delivery Data

21 of 35 records (60%) had no delivery time listed. 6 records had missing review counts. These gaps were disproportionately concentrated in newer or lower-volume outlets, which limits the completeness of operational efficiency analysis for that segment.

6.4 Cloud Kitchen Identification

Platforms do not consistently expose the restaurant type field in their API responses. Cloud kitchen classification was inferred from listing indicators (absence of dine-in address, "delivery only" markers, virtual brand signals) and manually cross-verified. This introduced subjectivity into the type field for approximately 4–5 records where classification was ambiguous.

7. Key Insights from Data

7.1 Cuisine Saturation Analysis

Individual cuisine tags were extracted from multi-tag strings and counted independently across all 35 restaurants. The resulting distribution revealed three clearly saturated segments:

Cuisine	Count	Status
Fast Food	7	⚠️ Saturated
North Indian	6	⚠️ Saturated
Continental	6	⚠️ Saturated
Italian	5	Moderate
South Indian	4	Moderate
Beverages	3	Moderate
Biryani	1	✓ Opportunity
Maharashtrian	1	✓ Opportunity
Middle Eastern	1	✓ Opportunity

Fast Food (7), North Indian (6), and Continental (6) together account for approximately 54% of all cuisine coverage on College Road. New entrants in these categories face direct competition from both established national chains and locally popular dine-in restaurants — a high-risk proposition at limited capital.

7.2 Underrepresented Cuisines and Market Gaps

Three cuisine categories represent clear market gaps based on low occurrence and high inherent demand:

- **Biryani (1 outlet):** One provider exists (Biryani Express Inn, rating: 3.4, reviews: 36). Biryani is among the highest-order-volume categories on Indian delivery platforms nationally. A single low-rated outlet represents minimal supply against strong latent demand.
- **Maharashtrian (1 outlet):** Nashik is a Maharashtrian city with cultural affinity for local cuisine. No restaurant currently leads with a Maharashtrian identity, representing an untapped local pride segment.
- **Middle Eastern (1 outlet):** Al Arabian Express (rating: 4.6, 16,000+ reviews) dominates this segment alone. While one strong competitor exists, the category's strength confirms demand for non-Indian formats.

7.3 Pricing Trends

The median cost-for-two across the 29 records with available pricing is approximately ₹400. Cloud kitchens tend to price at or slightly below the median, consistent with the delivery-only value proposition. The pricing range across all restaurant types is ₹150 (budget snack outlets) to ₹800 (premium dine-in and QSR combos).

7.4 Cloud Kitchen Opportunity

A Biryani-focused cloud kitchen targeting students and working professionals at ₹149–349 per item represents the strongest entry opportunity based on the data. The rationale:

- Zero quality benchmark to displace (existing Biryani outlet: 3.4 rating, 36 reviews)

- Cloud kitchen economics eliminate rent and service costs, enabling competitive pricing
- High-demand, low-supply category with no dominant brand in the locality
- Menu simplicity (3–5 SKUs) supports operational efficiency from launch

8. Limitations and Future Improvements

8.1 Dataset Size

A 35-record dataset, while sufficient for a pilot intelligence exercise, limits statistical confidence in cuisine frequency counts and pricing averages. A dataset of 100+ restaurants across multiple Nashik localities would enable comparative analysis and stronger market sizing conclusions.

8.2 Automation of Data Collection

Manual extraction is time-intensive and prone to human error during transcription. Future iterations should implement automated data pipelines using official platform APIs (where available) or structured web scraping using tools such as Python's Playwright or Selenium with rate-limiting and CAPTCHA management.

8.3 Sentiment Analysis

The current dataset captures review counts but not review content. Integrating sentiment analysis of customer reviews — using NLP libraries such as VADER or transformer-based models — would add a qualitative dimension to the quality signal beyond the aggregate star rating.

8.4 Temporal Coverage

Data was collected at a single point in time. Longitudinal data (weekly or monthly snapshots) would allow trend analysis — identifying emerging cuisine preferences, seasonal demand patterns, and rating trajectory for specific outlets.

8.5 Location-Based Analysis

Adding GPS coordinates or delivery zone data for each restaurant would enable heatmap-based gap analysis — identifying micro-zones within College Road that are underserved by existing outlets. This would significantly improve location selection precision for a cloud kitchen launch.

9. Conclusion

This project demonstrates the end-to-end application of data collection, cleaning, relational database design, and business intelligence analysis to a real-world market context. Starting from raw, inconsistently formatted platform data and arriving at a structured, query-ready dataset with actionable strategic insights reflects the core value proposition of applied data analytics.

The key output of this exercise — a recommendation to enter the College Road, Nashik market with a Biryani-focused cloud kitchen at a ₹149–349 price point — is grounded in empirical evidence:

cuisine frequency analysis, competitive gap identification, pricing benchmarks, and operational efficiency benchmarks drawn directly from the dataset.

More broadly, this project illustrates that structured data analysis, even at modest scale, can meaningfully reduce decision risk in a capital-constrained environment such as a first-time cloud kitchen launch. As the Indian food delivery market continues to grow, intelligence-driven market entry will increasingly define the difference between outlets that achieve sustainable volume and those that exit within their first operating year.

End of Report